Generative AI Interview Questions

Section 1: Generative AI Fundamentals

1. What is Generative AI, and how does it differ from traditional Artificial Intelligence?
2. What is the difference between Generative AI and discriminative AI?
3. What types of content can Generative AI systems produce?
4. What is the difference between Artificial Intelligence, Machine Learning, Deep Learning and Generative AI?
5. What is a generative model, and how does it learn patterns from training data?
6. What are the major types of generative models?
7. What is the difference between rule-based systems and generative models?
8. What are the common real-world applications of Generative AI?
9. What are the main limitations and risks of Generative AI?
10. What is the difference between open-source, open-weight and proprietary AI models?

Section 2: Large Language Model Fundamentals

11. What is a Large Language Model (LLM)?
12. How does an LLM learn to generate human-like text?
13. What is next-token prediction?
14. What are model parameters, and how do they affect an LLM's capabilities?
15. What is the difference between model parameters and hyperparameters?
16. What is a context window, and how does it limit the information a model can process?
17. What is the difference between training, fine-tuning and inference?
18. What is the difference between a base model and an instruction-tuned model?
19. What is the difference between a language model and a chat model?

20. Why can an LLM generate incorrect information even when its response sounds convincing?

Section 3: Transformer Architecture

21. What is the Transformer architecture, and why is it important in Generative AI?
22. What is the difference between an encoder, a decoder and an encoder–decoder architecture?
23. What is self-attention, and how does it help a model understand context?
24. What are query, key and value vectors in the attention mechanism?
25. What is scaled dot-product attention?
26. What is multi-head attention, and why is it useful?
27. What is positional encoding, and why is positional information important?
28. What is causal masking, and why is it used in GPT-style models?
29. What is the difference between BERT-style encoder models and GPT-style decoder models?
30. Why can standard Transformer attention become expensive when processing long sequences?

Section 4: Prompt Engineering

31. What is prompt engineering, and why is it important?
32. What is the difference between zero-shot, one-shot and few-shot prompting?
33. What is chain-of-thought prompting, and when might reasoning-oriented prompting be useful?
34. What is role prompting, and how can it influence a model's response?
35. What is the difference between system, developer and user instructions in a chat-based model API?
36. What is structured-output prompting, and how can you request valid JSON?
37. What is prompt chaining, and how does it differ from a single prompt?
38. What is prompt templating, and how can it improve consistency in an application?
39. How do temperature, top-k and top-p affect text generation?

40. How would you systematically test and improve a prompt that produces inconsistent answers?

Section 5: Tokens, Embeddings and Semantic Search

41. What is a token, and how does tokenisation affect LLM input and output?
42. Why does a word not always correspond to a single token?
43. What is a token budget, and how would you manage it in an LLM application?
44. What are embeddings, and how do they represent the meaning of text?
45. What is the difference between word embeddings and sentence embeddings?
46. What is cosine similarity, and how is it used to compare embeddings?
47. What is a vector database, and why is it useful in Generative AI applications?
48. What is the difference between keyword search and semantic search?
49. What is the difference between sparse and dense retrieval?
50. How would you build a semantic search system for a collection of company documents?

Section 6: Retrieval-Augmented Generation (RAG) Fundamentals

51. What is Retrieval-Augmented Generation (RAG), and why is it used?
52. What problems does RAG solve that a standalone LLM may struggle with?
53. What are the main components of a RAG architecture?
54. What happens during document ingestion in a RAG pipeline?
55. What is document chunking, and why is it necessary?
56. How do chunk size and chunk overlap affect retrieval quality?
57. What is the difference between indexing time and query time in RAG?
58. How does a retriever select relevant documents for a user query?
59. How does retrieved context help an LLM generate a grounded answer?
60. What are the main limitations of a basic RAG system?

Section 7: Advanced RAG and Retrieval Optimisation

61. What is hybrid search, and how does it combine lexical and semantic retrieval?
62. What is a reranker, and how does it improve the relevance of retrieved documents?
63. What is query rewriting, and why can it improve retrieval?
64. What is query expansion, and how does it differ from query rewriting?
65. What is metadata filtering, and when would you use it in a vector database?
66. What is parent-document retrieval, and how can it help balance retrieval precision and context?
67. What is multi-query retrieval, and when is it useful?
68. What is contextual compression in RAG?
69. What is the difference between basic RAG, agentic RAG and graph-based RAG?
70. How would you troubleshoot a RAG system that retrieves irrelevant documents despite having good embeddings?

Section 8: Fine-Tuning and Model Adaptation

71. What is fine-tuning, and when is it appropriate for an LLM?
72. What is the difference between full fine-tuning and parameter-efficient fine-tuning?
73. What is Low-Rank Adaptation (LoRA)?
74. What is QLoRA, and how does it reduce fine-tuning memory requirements?
75. What is supervised fine-tuning (SFT)?
76. What is instruction tuning, and how does it differ from domain-specific fine-tuning?
77. What is catastrophic forgetting during fine-tuning?
78. How would you prepare and validate a dataset for fine-tuning an LLM?
79. How would you decide between prompt engineering, RAG and fine-tuning for a business use case?
80. How would you evaluate whether a fine-tuned model has genuinely improved over the base model?

Section 9: Generative Models Beyond Text

81. What is a Generative Adversarial Network (GAN), and how does it generate synthetic data?
82. What are the generator and discriminator in a GAN?
83. What is a Variational Autoencoder (VAE)?
84. What is a diffusion model, and how does it generate images?
85. What is the difference between GANs, VAEs and diffusion models?
86. What is latent space in a generative model?
87. What is a multimodal AI model?
88. How can an AI model combine text and image inputs?
89. What are text-to-image and image-to-text generation?
90. How would you evaluate the quality of generated images, audio or video?

Section 10: LLM Application Development and Frameworks

91. How would you call an LLM using a Python API?
92. What is the difference between a synchronous and an asynchronous LLM call?
93. What are system prompts, message history and conversation state in a chatbot?
94. How would you maintain conversation history without exceeding the model's context window?
95. What is LangChain, and what problems does it help solve?
96. What is LangGraph, and when would you use it instead of a simple sequential chain?
97. What is the difference between using an LLM provider's SDK directly and using an orchestration framework?
98. How would you implement structured output validation using Python and Pydantic?
99. How would you handle API errors, timeouts, rate limits and retries in an LLM application?
100. How would you design a modular Python application that supports multiple LLM providers?

Section 11: AI Agents and Tool Calling

101. What is an AI agent, and how does it differ from a conventional chatbot?

102. What is tool calling, and how can an LLM use external functions?
103. What is the difference between tool calling and retrieving information through RAG?
104. What are the plan, act and observe stages of an agent workflow?
105. What is the difference between a fixed workflow and an autonomous agent?
106. What is an agent's state, and why is state management important?
107. What are memory and context in an AI agent, and how do they differ?
108. What are task delegation and multi-agent collaboration?
109. What are the risks of giving an agent unrestricted access to external tools?
110. How would you design an agent that searches documents, calls an external API and produces a validated final report?

Section 12: Evaluation and Reliability

111. Why is evaluating Generative AI systems more difficult than evaluating conventional classification models?
112. What is the difference between evaluating an LLM, a RAG pipeline and a complete AI application?
113. What are perplexity and next-token prediction loss, and what do they measure?
114. What are BLEU and ROUGE, and what are their limitations when evaluating generated text?
115. What is LLM-as-a-judge evaluation, and what biases can it introduce?
116. How would you measure factual correctness and faithfulness in a RAG-generated answer?
117. What is retrieval recall, and how does it affect RAG performance?
118. What is an evaluation dataset or benchmark, and how would you create one for a domain-specific chatbot?
119. How would you test an LLM application for consistency, hallucinations and regression after a model update?
120. What is the difference between offline evaluation and online evaluation?

Section 13: Safety, Security and Responsible AI

121. What is hallucination in Generative AI, and what techniques can reduce it?

122. What is prompt injection, and how can it affect an LLM application?
123. What is the difference between direct and indirect prompt injection?
124. What is data poisoning, and how can it affect model training or retrieval systems?
125. What is sensitive-data leakage, and how can it occur in an LLM application?
126. What are guardrails, and how can they enforce input and output constraints?
127. Why should an application validate LLM-generated JSON instead of trusting the model's output?
128. What is the principle of least privilege, and how does it apply to AI agents?
129. How would you prevent a RAG system from returning documents that a user is not authorised to access?
130. What responsible AI considerations apply to bias, privacy, transparency and human oversight?

Section 14: Deployment, Optimisation and Production

131. What is inference latency, and which factors influence it in an LLM application?
132. What is quantisation, and how can it reduce model memory requirements?
133. What is the difference between CPU and GPU inference for LLMs?
134. What is batching, and how can it improve inference throughput?
135. What is the difference between throughput and latency?
136. What is prompt caching, and when can it reduce cost or latency?
137. What is streaming generation, and how does it improve the user experience?
138. How would you estimate the token cost of an LLM application before deploying it?
139. How would you monitor token usage, API costs, errors, latency and output quality in production?
140. How would you deploy an LLM application using FastAPI, Docker and a cloud platform?